

Michele Bilko

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EDUCATION

Tufts University, Medford, MA

Sep 2026 – May 2027

M.S. in Computer Science · Incoming Fall 2026

Boston University, Boston, MA

Sep 2024 – May 2026

B.A. in Computer Science, Minor in Philosophy · Dean's List

Relevant Coursework: Advanced Algorithms, Theory of Computation, Fine-Grained Complexity, Image & Video Computing, Database Systems, Software Engineering, Full-Stack Development, Probability in Computing

University of California, Santa Cruz, CA

Sep 2023 – Jun 2024

B.S. coursework in Computer Science & Game Development (transferred) · Dean's Honors List

RESEARCH EXPERIENCE

Directed Study in Graph Algorithms — Boston University

Jan 2026 – Present

Advised by Prof. Dora Erdős

- Investigating graph conductance and open problems in network connectivity; built a Python/NetworkX framework benchmarking distributed conductance-testing algorithms (BTT vs. Fichtenberger–Vasudev) across varied graph topologies.

Satellite Image Restoration Research — Boston University

Jan 2026 – May 2026

- Built a deep-learning pipeline reconstructing ~22% of stratified data gaps in Landsat 7 imagery from the 2003 Scan Line Corrector failure; extended the LaMa inpainting model with multi-temporal reference guidance and CBAM attention. Paper in preparation.

PROFESSIONAL EXPERIENCE

Bike Shop Technician — REI, Boston, MA

May 2026 – Aug 2026

- Diagnosed mechanical issues on the spot for a steady stream of customer bikes, providing accurate cost and time estimates and prioritizing repairs to keep the shop moving.
- Managed daily shop operations including intake, service tracking, and customer communication in a fast-paced, seasonal-peak retail environment.

Intern — Columbia University Media Center for Art History

Sep 2022 – Jun 2023

- Contributed to the design and ongoing maintenance of the Media Center's public-facing website: learn.columbia.edu/media-center-art-history.
- Engineered an automated data-processing pipeline (Java, Python, pandas) transforming 20,000+ lines of imaging metadata into a normalized, searchable schema for digital-humanities research infrastructure.
- Applied photogrammetry and 3D-reconstruction algorithms using Structure-from-Motion (SfM) pipelines to build virtual tours.

TECHNICAL PROJECTS

Central Rock Gym Route Tracking System · React, TypeScript, Django, PostgreSQL

Aug 2025 – Present

- Built a full-stack web application end-to-end with a React/TypeScript frontend and Django backend (custom ORM queries and triggers), integrating real-time data synchronization and a D3.js analytics dashboard over a production SQL database.

PUBLICATIONS & WRITING

- Authored the Wikipedia article on Fine-Grained Complexity, synthesizing core results for a broad technical audience.
- Graduate-level survey paper on fine-grained complexity (CS 530); two research papers in preparation (Landsat 7 inpainting; graph conductance-testing comparison).

ACTIVITIES

- U.S. Cyber Challenge — Top 50 Nationally (2025); active member of the BU Cybersecurity Association, competing in CTF competitions across web exploitation, reverse engineering, and forensics.

TECHNICAL SKILLS

Languages: Python, C++, SQL, Java, C#, JavaScript/TypeScript, HTML/CSS

Frameworks & Libraries: React, Django, Node.js, Express, Flask, PyTorch, TensorFlow, scikit-learn, NumPy, pandas, SQLAlchemy, NetworkX, D3.js

Tools & Practices: Git/GitHub, Docker, Linux/Unix, LaTeX, PostgreSQL, RESTful API design, Agile, unit & integration testing, web scraping (requests, BeautifulSoup, Selenium)

Focus Areas: Full-Stack Development, Machine Learning & Computer Vision, Algorithms & Complexity, Database Systems

AI Tools: Productive with LLM-based coding assistants and agents for prototyping and task automation.